

Scientific Hypothesis Generation Report

[Phenomenon Name]

Evidence-Based Competing Hypotheses

August 26, 2026

Executive Summary

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Phenomenon: [One paragraph: What was observed? Why is it important?]

Key Question: [Single sentence stating the central question]

Competing Hypotheses:

1. **[H1 Title]:** [One sentence mechanistic summary]
2. **[H2 Title]:** [One sentence mechanistic summary]
3. **[H3 Title]:** [One sentence mechanistic summary]
4. **[Add H4 & H5 if applicable]**

Recommended Approach: [One sentence on priority experiments]

1 Competing Hypotheses

This section presents [3-5] distinct mechanistic hypotheses. Detailed literature review and comprehensive evidence are in Appendix A.

Hypothesis 1: [Concise Descriptive Title]

Hypothesis 1: [Title]

]

Mechanistic Explanation:

[Provide a BRIEF mechanistic explanation (1-2 paragraphs) of HOW and WHY. Keep concise - main text is limited to 4 pages total. Include only the essential mechanism. All detailed explanations go to Appendix A.

Example: "This hypothesis proposes that [mechanism X] operates through [pathway Y], resulting in [outcome Z]. The process initiates when [trigger], activating [component A] and ultimately producing the observed [phenomenon] (?)."]

Key Supporting Evidence:

evidence point 1 (?)

evidence point 2 (?)

evidence point 3 (?)

Core Assumptions:

critical assumption 1

critical assumption 2

Hypothesis 2: [Concise Descriptive Title]

Hypothesis 2: [Title]

]

Mechanistic Explanation:

[BRIEF mechanistic explanation (1-2 paragraphs) distinct from Hypothesis 1. Keep concise.]

Key Supporting Evidence:

point 1 with citation

point 2 with citation

point 3 with citation

Core Assumptions:

Critical assumption 1

Critical assumption 2

Hypothesis 3: [Concise Descriptive Title]

Hypothesis 3: [Title]

]

Mechanistic Explanation:

[BRIEF mechanistic explanation (1-2 paragraphs) distinct from previous hypotheses.]

Key Supporting Evidence:

point 1 with citation

point 2 with citation

point 3 with citation

Core Assumptions:

Critical assumption 1

Critical assumption 2

2 Testable Predictions

Key predictions from each hypothesis. Full prediction details and additional predictions in Appendix B.

Predictions from Hypothesis 1

Predictions: Hypothesis 1

Prediction 1.1: [Most critical prediction]

- **Expected Outcome:** [Specific result with magnitude if possible]
- **Falsification:** [What would disprove it]

Prediction 1.2: [Second most critical prediction]

- **Expected Outcome:** [Specific result]
- **Falsification:** [What would disprove it]

Predictions from Hypothesis 2

Predictions: Hypothesis 2

Prediction 2.1: [Most critical prediction]

- **Expected Outcome:** [Specific result]
- **Falsification:** [What would disprove it]

Prediction 2.2: [Second most critical prediction]

- **Expected Outcome:** [Specific result]
- **Falsification:** [What would disprove it]

Predictions from Hypothesis 3

Predictions: Hypothesis 3

[1-2 most critical predictions only, following same brief structure]

3 Critical Comparisons

Highest priority comparison for distinguishing hypotheses. Additional comparisons in Appendix B.

Priority Comparison: Hypothesis 1 vs. Hypothesis 2

H1 vs. H2: Key Distinction

Fundamental Difference: [One sentence on core mechanistic difference]

Discriminating Experiment: [Brief description of key experiment to distinguish them]

Outcome Interpretation:

- If [Result A]: H1 supported
- If [Result B]: H2 supported

Highest Priority Test: [Name of single most important experiment]

Justification: [2-3 sentences on why this is highest priority considering informativeness and feasibility. Full experimental details in Appendix B.]

Appendix A: Comprehensive Literature Review

This appendix provides detailed synthesis of existing literature, extensive background context, and comprehensive citations supporting the hypotheses presented in this report.

A.1 Phenomenon Background and Context

[Provide extensive background on the phenomenon. This section should be comprehensive, including:

- Historical context and when the phenomenon was first observed
- Detailed description of what is known about the phenomenon
- Why this phenomenon is scientifically important
- Practical or clinical implications if applicable
- Current debates or controversies in the field

Include extensive citations throughout. Aim for 10-15 citations in this subsection alone.]

A.2 Current Understanding and Established Mechanisms

[Synthesize what is currently understood about this phenomenon:

- Established theories or frameworks that may apply
- Known mechanisms from related systems or analogous phenomena
- Molecular, cellular, or systemic processes that are well-characterized
- Population-level patterns that have been documented
- Computational or theoretical models that have been proposed

Include 15-20 citations covering recent reviews, primary research papers, and foundational studies.]

A.3 Evidence Supporting Hypothesis 1

[Provide detailed discussion of all evidence supporting Hypothesis 1. This goes beyond the brief bullet points in the main text:

- Detailed findings from key papers
- Mechanistic studies showing relevant pathways
- Data from analogous systems
- Theoretical support
- Any preliminary or indirect evidence

Include 8-12 citations specific to this hypothesis.]

A.4 Evidence Supporting Hypothesis 2

[Same structure as A.3, focused on Hypothesis 2. Include 8-12 citations.]

A.5 Evidence Supporting Hypothesis 3

[Same structure as A.3, focused on Hypothesis 3. Include 8-12 citations.]

A.6 Conflicting Findings and Unresolved Debates

[Discuss contradictions in the literature:

- Studies with conflicting results
- Ongoing debates about mechanisms
- Alternative interpretations of existing data
- Methodological issues that complicate interpretation
- Areas where consensus has not been reached

Include 5-10 citations highlighting key controversies.]

A.7 Knowledge Gaps and Limitations

[Identify what is still unknown:

- Aspects of the phenomenon that lack clear explanation
- Missing data or unstudied conditions
- Limitations of current methods or approaches
- Questions that remain unanswered
- Assumptions that have not been tested

Include 3-5 citations discussing limitations or identifying gaps.]

Appendix B: Detailed Experimental Designs

This appendix provides comprehensive experimental protocols for testing each hypothesis, including methods, controls, sample sizes, statistical approaches, and feasibility assessments.

B.1 Experiments for Testing Hypothesis 1

Experiment 1A: [Descriptive Title]

Design Type: [e.g., In vitro dose-response / In vivo knockout / Clinical RCT / Observational cohort / Computational model]

Objective: [What specific aspect of Hypothesis 1 does this experiment test? What question does it answer?]

Detailed Methods:

- **System/Model:** [What system, organism, cell type, or population will be studied? Include species, strains, patient populations, etc.]
- **Intervention/Manipulation:** [What will be varied or manipulated? Include specific treatments, genetic modifications, interventions, etc.]
- **Measurements:** [What outcomes will be measured? Include primary and secondary endpoints, measurement techniques, timing of measurements]
- **Controls:** [What control conditions will be included? Negative controls, positive controls, vehicle controls, sham procedures, etc.]
- **Sample Size:** [Estimated n per group with power analysis justification if possible. Include assumptions about effect size and variability.]
- **Randomization & Blinding:** [How will subjects be randomized? Who will be blinded?]
- **Statistical Analysis:** [Specific statistical tests planned, correction for multiple comparisons, significance thresholds]

Expected Timeline: [Rough estimate of duration from start to completion]

Resource Requirements:

- **Equipment:** [Specialized equipment needed]
- **Materials:** [Key reagents, animals, human subjects]
- **Expertise:** [Specialized skills or training required]
- **Estimated Cost:** [Rough cost estimate if applicable]

Feasibility Assessment: [High/Medium/Low with justification. Consider technical challenges, resource availability, ethical considerations]

Potential Confounds and Mitigation:

how to address it
how to address it

how to address it

Experiment 1B: [Alternative or Complementary Approach]

[Follow same detailed structure as Experiment 1A. This should be an alternative method to test the same aspect of Hypothesis 1, or a complementary experiment that tests a different aspect.]

B.2 Experiments for Testing Hypothesis 2**Experiment 2A: [Descriptive Title]**

[Follow same detailed structure as above]

Experiment 2B: [Alternative or Complementary Approach]

[Follow same detailed structure as above]

B.3 Experiments for Testing Hypothesis 3

[Continue with same structure for all hypotheses]

B.4 Discriminating Experiments

[Provide detailed protocols for the priority experiments identified in Section 4 that distinguish between hypotheses]

Appendix C: Quality Assessment

This appendix provides detailed evaluation of each hypothesis against established quality criteria.

C.1 Comparative Quality Assessment

Criterion	Hypothesis 1	Hypothesis 2	Hypothesis 3
Testability	[Strong/Moderate/Weak] [Brief note: why?]	[Rating & note]	[Rating & note]
Falsifiability	[Rating & note]	[Rating & note]	[Rating & note]
Parsimony	[Rating & note]	[Rating & note]	[Rating & note]
Explanatory Power	[Rating & note]	[Rating & note]	[Rating & note]
Scope	[Rating & note]	[Rating & note]	[Rating & note]
Consistency	[Rating & note]	[Rating & note]	[Rating & note]
Novelty	[Rating & note]	[Rating & note]	[Rating & note]

Table 1: Comparative assessment of hypotheses across quality criteria. Strong = meets criterion very well; Moderate = partially meets criterion; Weak = does not meet criterion well.

C.2 Detailed Evaluation: Hypothesis 1

Strengths:

is advantageous
specific strength 2
specific strength 3
as applicable

Weaknesses:

of the limitation
specific weakness 2
specific weakness 3
as applicable

Overall Assessment:

[Provide a comprehensive 1-2 paragraph assessment of Hypothesis 1’s quality and viability. Consider:

- How well does it balance the various quality criteria?
- What are the key trade-offs?

- Under what conditions would this be the most promising hypothesis?
- What are the major challenges to testing or validating it?
- How does it compare overall to competing hypotheses?

]

C.3 Detailed Evaluation: Hypothesis 2

[Follow same structure as C.2]

C.4 Detailed Evaluation: Hypothesis 3

[Follow same structure as C.2]

C.5 Recommendations Based on Quality Assessment

[Synthesize the quality assessments to provide recommendations:

- Which hypothesis appears most promising overall?
- Which hypothesis should be tested first? Why?
- Are there scenarios where different hypotheses would be preferred?
- Could multiple hypotheses be partially correct?
- What would need to be true for each hypothesis to be viable?

]

Appendix D: Supplementary Evidence

This appendix provides additional supporting information, including analogous mechanisms, relevant data, and context that further informs the hypotheses.

D.1 Analogous Mechanisms in Related Systems

[Discuss similar mechanisms or phenomena in related systems that provide insight:

- How do analogous systems behave?
- What mechanisms operate in those systems?
- How might lessons from related systems apply here?
- What similarities and differences exist?

Include citations to relevant comparative studies.]

D.2 Preliminary Data or Observations

[If applicable, discuss any preliminary data, pilot studies, or anecdotal observations that informed hypothesis generation but weren't formally published or well-documented.]

D.3 Theoretical Frameworks

[Discuss broader theoretical frameworks that relate to the hypotheses:

- What general principles or theories apply?
- How do the hypotheses fit within established frameworks?
- Are there mathematical or computational models that support any hypothesis?

]

D.4 Historical Context and Evolution of Ideas

[Provide historical perspective on how thinking about this phenomenon has evolved, what previous hypotheses have been proposed and tested, and what lessons have been learned from past attempts to explain the phenomenon.]